Data Protection

Fundamentals of Cloud Security, First Edition - Lab 05

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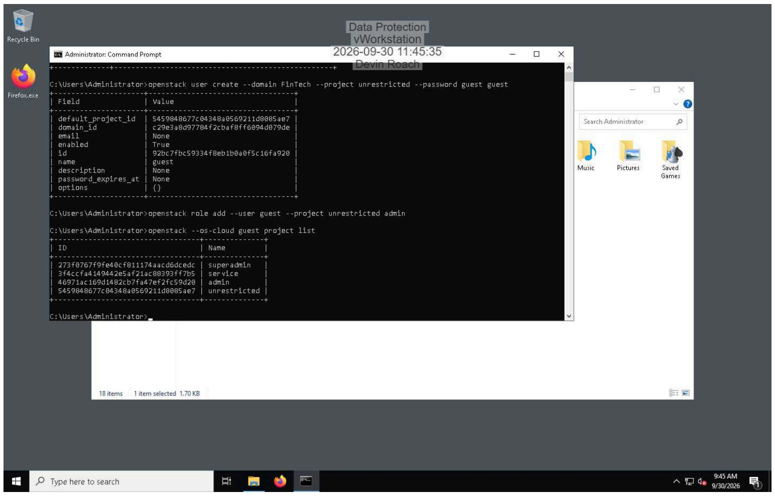
Progress: 100%

Section 1: Hands-On Demonstration

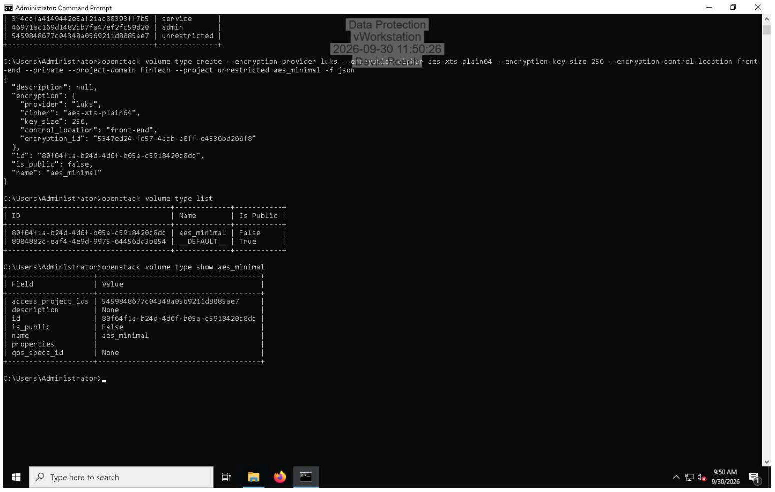
Part 1: Configuring Volume Encryption with Cinder

12. **Make a screen capture** showing the **guest user project list command and results**.



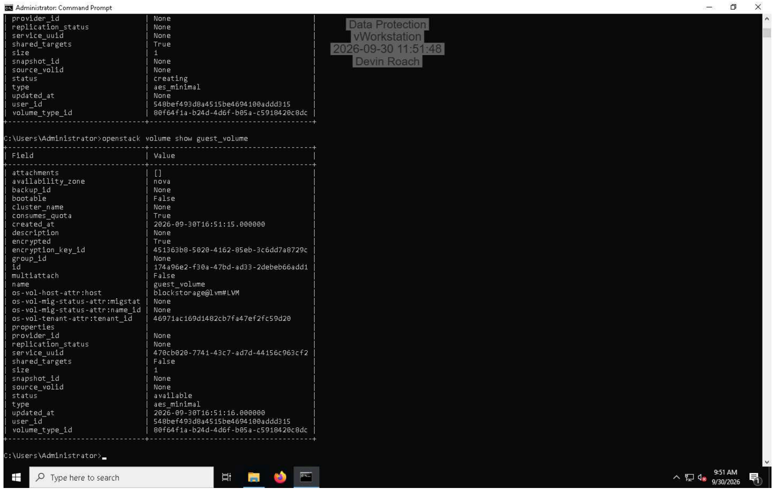
```
+---------------------+----------------------------------------------------+

C:\Users\Administrator>openstack user create --domain FinTech --project unrestricted --password guest guest
+---------------------+----------------------------------+
| Field               | Value                            |
+---------------------+----------------------------------+
| default_project_id  | 5459848677c04348a0569211d8085ae7 |
| domain_id           | c29e3a8d97784f2cbaf8ff6094d079de |
| email               | None                             |
| enabled             | True                             |
| id                  | 92bc7fbc59334f8eb1b0a0f5c16fa920 |
| name                | guest                            |
| description         | None                             |
| password_expires_at | None                             |
| options             | {}                               |
+---------------------+----------------------------------+

C:\Users\Administrator>openstack role add --user guest --project unrestricted admin

C:\Users\Administrator>openstack --os-cloud guest project list
+----------------------------------+--------------+
| ID                               | Name         |
+----------------------------------+--------------+
| 273f0767f9fe40cf811174aacd6dcedc | superadmin   |
| 3f4ccfa4149442e5af21ac88393ff7b5 | service      |
| 46971ac169d1482cb7fa47ef2fc59d20 | admin        |
| 5459848677c04348a0569211d8085ae7 | unrestricted |
+----------------------------------+--------------+

C:\Users\Administrator>
```

16. **Make a screen capture** showing the **details of the aes_minimal volume type**.



```
| 3f4ccfa4149442e5af21ac88393ff7b5 | service      |
| 46971ac169d1482cb7fa47ef2fc59d20 | admin        |
| 5459848677c04348a0569211d8085ae7 | unrestricted |
+----------------------------------+--------------+

C:\Users\Administrator>openstack volume type create --encryption-provider luks --encryption-cipher aes-xts-plain64 --encryption-key-size 256 --encryption-control-location front
-end --private --project-domain FinTech --project unrestricted aes_minimal -f json
{
  "description": null,
  "encryption": {
    "provider": "luks",
    "cipher": "aes-xts-plain64",
    "key_size": 256,
    "control_location": "front-end",
    "encryption_id": "5347ed24-fc57-4acb-a0ff-e4536bd266f8"
  },
  "id": "80f64f1a-b24d-4d6f-b05a-c5918420c8dc",
  "is_public": false,
  "name": "aes_minimal"
}

C:\Users\Administrator>openstack volume type list
+--------------------------------------+-------------+-----------+
| ID                                   | Name        | Is Public |
+--------------------------------------+-------------+-----------+
| 80f64f1a-b24d-4d6f-b05a-c5918420c8dc | aes_minimal | False     |
| 8904882c-eaf4-4e9d-9975-64456dd3b054 | __DEFAULT__ | True      |
+--------------------------------------+-------------+-----------+

C:\Users\Administrator>openstack volume type show aes_minimal
+--------------------+--------------------------------------+
| Field              | Value                                |
+--------------------+--------------------------------------+
| access_project_ids | 5459848677c04348a0569211d8085ae7     |
| description        | None                                 |
| id                 | 80f64f1a-b24d-4d6f-b05a-c5918420c8dc |
| is_public          | False                                |
| name               | aes_minimal                          |
| properties         |                                      |
| qos_specs_id       | None                                 |
+--------------------+--------------------------------------+

C:\Users\Administrator>
```

20. **Make a screen capture** showing the **details of the encrypted guest volume with a status of available**.

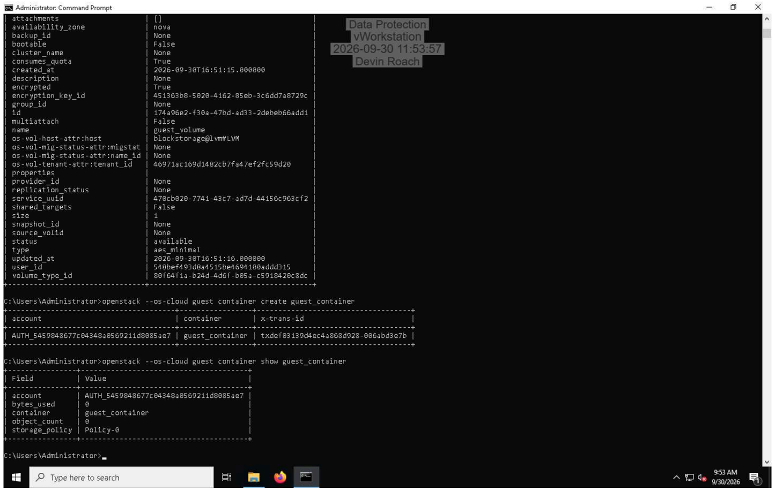


Data Protection
vWorkstation
2026-09-30 11:51:48
Devin Roach

```
| provider_id                    | None                                 |
| replication_status             | None                                 |
| service_uuid                   | None                                 |
| shared_targets                 | True                                 |
| size                           | 1                                    |
| snapshot_id                    | None                                 |
| source_volid                   | None                                 |
| status                         | creating                             |
| type                           | aes_minimal                          |
| updated_at                     | None                                 |
| user_id                        | 548bef493d8a4515be4694100addd315     |
| volume_type_id                 | 80f64f1a-b24d-4d6f-b05a-c5918420c8dc |
+--------------------------------+--------------------------------------+

C:\Users\Administrator>openstack volume show guest_volume
+--------------------------------+--------------------------------------+
| Field                          | Value                                |
+--------------------------------+--------------------------------------+
| attachments                    | []                                   |
| availability_zone              | nova                                 |
| backup_id                      | None                                 |
| bootable                       | False                                |
| cluster_name                   | None                                 |
| consumes_quota                 | True                                 |
| created_at                     | 2026-09-30T16:51:15.000000           |
| description                    | None                                 |
| encrypted                      | True                                 |
| encryption_key_id              | 451363b8-5020-4162-85eb-3c6dd7a8729c |
| group_id                       | None                                 |
| id                             | 174a96e2-f30a-47bd-ad33-2debeb66add1 |
| multiattach                    | False                                |
| name                           | guest_volume                         |
| os-vol-host-attr:host          | blockstorage@lvm#LVM                 |
| os-vol-mig-status-attr:migstat | None                                 |
| os-vol-mig-status-attr:name_id | None                                 |
| os-vol-tenant-attr:tenant_id   | 46971ac169d1482cb7fa47ef2fc59d20     |
| properties                     |                                      |
| provider_id                    | None                                 |
| replication_status             | None                                 |
| service_uuid                   | 470cb020-7741-43c7-ad7d-44156c963cf2 |
| shared_targets                 | False                                |
| size                           | 1                                    |
| snapshot_id                    | None                                 |
| source_volid                   | None                                 |
| status                         | available                            |
| type                           | aes_minimal                          |
| updated_at                     | 2026-09-30T16:51:16.000000           |
| user_id                        | 548bef493d8a4515be4694100addd315     |
| volume_type_id                 | 80f64f1a-b24d-4d6f-b05a-c5918420c8dc |
+--------------------------------+--------------------------------------+

C:\Users\Administrator>_
```

Part 2: Managing Data with Object Storage

3. **Make a screen capture** showing the **details for the guest container**.



Data Protection
vWorkstation
2026-09-30 11:53:57
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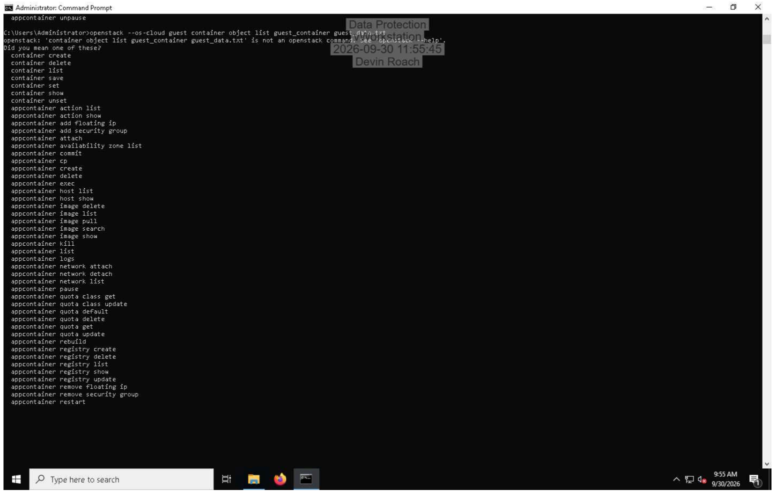
```
| attachments                    | []                                   |
| availability_zone              | nova                                 |
| backup_id                      | None                                 |
| bootable                       | False                                |
| cluster_name                   | None                                 |
| consumes_quota                 | True                                 |
| created_at                     | 2026-09-30T16:51:15.000000           |
| description                    | None                                 |
| encrypted                      | True                                 |
| encryption_key_id              | 451363b8-5020-4162-85eb-3c6dd7a8729c |
| group_id                       | None                                 |
| id                             | 174a96e2-f30a-47bd-ad33-2debeb66add1 |
| multiattach                    | False                                |
| name                           | guest_volume                         |
| os-vol-host-attr:host          | blockstorage@lvm#LVM                 |
| os-vol-mig-status-attr:migstat | None                                 |
| os-vol-mig-status-attr:name_id | None                                 |
| os-vol-tenant-attr:tenant_id   | 46971ac169d1482cb7fa47ef2fc59d20     |
| properties                     |                                      |
| provider_id                    | None                                 |
| replication_status             | None                                 |
| service_uuid                   | 470cb020-7741-43c7-ad7d-44156c963cf2 |
| shared_targets                 | False                                |
| size                           | 1                                    |
| snapshot_id                    | None                                 |
| source_volid                   | None                                 |
| status                         | available                            |
| type                           | aes_minimal                          |
| updated_at                     | 2026-09-30T16:51:16.000000           |
| user_id                        | 548bef493d8a4515be4694100addd315     |
| volume_type_id                 | 80f64f1a-b24d-4d6f-b05a-c5918420c8dc |
+--------------------------------+--------------------------------------+

C:\Users\Administrator>openstack --os-cloud guest container create guest_container
+---------------------------------------+-----------------+----------------------------------------+
| account                               | container       | x-trans-id                             |
+---------------------------------------+-----------------+----------------------------------------+
| AUTH_5459848677c04348a0569211d8085ae7 | guest_container | txdef03139d4ec4a868d928-006abd3e7b |
+---------------------------------------+-----------------+----------------------------------------+

C:\Users\Administrator>openstack --os-cloud guest container show guest_container
+----------------+---------------------------------------+
| Field          | Value                                 |
+----------------+---------------------------------------+
| account        | AUTH_5459848677c04348a0569211d8085ae7 |
| bytes_used     | 0                                     |
| container      | guest_container                       |
| object_count   | 0                                     |
| storage_policy | Policy-0                              |
+----------------+---------------------------------------+

C:\Users\Administrator>_
```

7. **Make a screen capture** showing the **details for the guest_data.txt object**.



Data Protection
vWorkstation
2026-09-30 11:55:45
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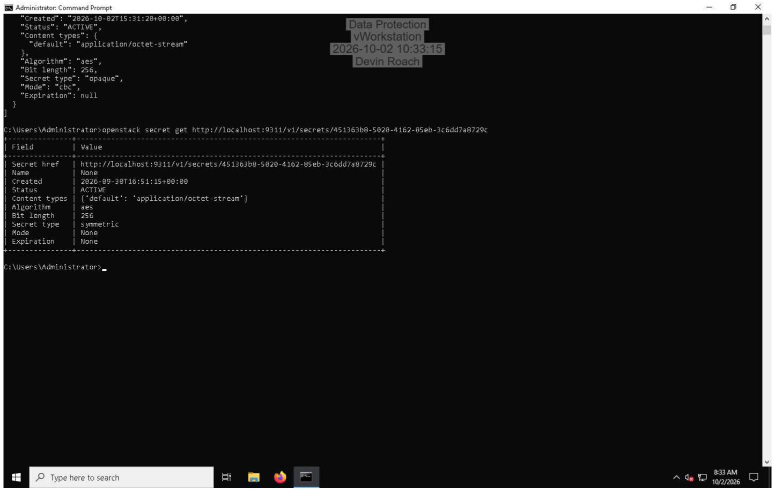
```
  appcontainer unpause

C:\Users\Administrator>openstack --os-cloud guest container object list guest_container guest_data.txt
openstack: 'container object list guest_container guest_data.txt' is not an openstack command. See 'openstack --help'.
Did you mean one of these?
  container create
  container delete
  container list
  container save
  container set
  container show
  container unset
  appcontainer action list
  appcontainer action show
  appcontainer add floating ip
  appcontainer add security group
  appcontainer attach
  appcontainer availability zone list
  appcontainer commit
  appcontainer cp
  appcontainer create
  appcontainer delete
  appcontainer exec
  appcontainer host list
  appcontainer host show
  appcontainer image delete
  appcontainer image list
  appcontainer image pull
  appcontainer image search
  appcontainer image show
  appcontainer kill
  appcontainer list
  appcontainer logs
  appcontainer network attach
  appcontainer network detach
  appcontainer network list
  appcontainer pause
  appcontainer quota class get
  appcontainer quota class update
  appcontainer quota default
  appcontainer quota delete
  appcontainer quota get
  appcontainer quota update
  appcontainer rebuild
  appcontainer registry create
  appcontainer registry delete
  appcontainer registry list
  appcontainer registry show
  appcontainer registry update
  appcontainer remove floating ip
  appcontainer remove security group
  appcontainer restart
```

Part 3: Storing Secrets with Barbican

5. **Make a screen capture** showing the **newly created secret details**.

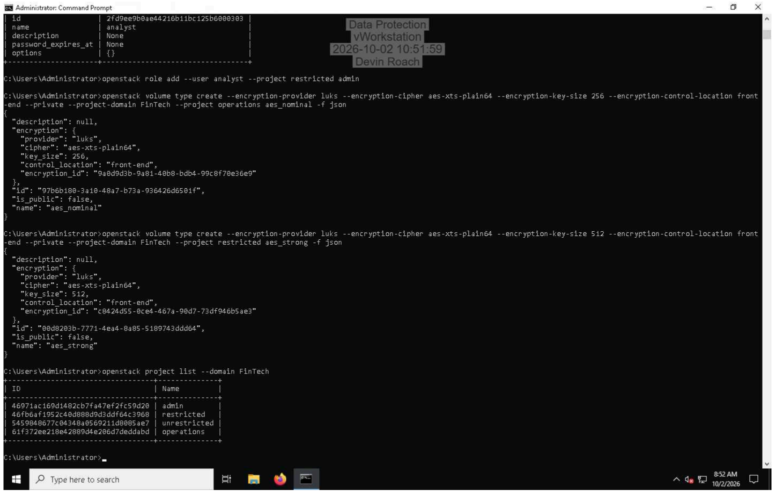


```
    "Created": "2026-10-02T15:31:20+00:00",
    "Status": "ACTIVE",
    "Content types": {
      "default": "application/octet-stream"
    },
    "Algorithm": "aes",
    "Bit length": 256,
    "Secret type": "opaque",
    "Mode": "cbc",
    "Expiration": null
  }
]

C:\Users\Administrator>openstack secret get http://localhost:9311/v1/secrets/451363b8-5020-4162-85eb-3c6dd7a8729c
+---------------+-----------------------------------------------------------------------+
| Field         | Value                                                                 |
+---------------+-----------------------------------------------------------------------+
| Secret href   | http://localhost:9311/v1/secrets/451363b8-5020-4162-85eb-3c6dd7a8729c |
| Name          | None                                                                  |
| Created       | 2026-09-30T16:51:15+00:00                                             |
| Status        | ACTIVE                                                                |
| Content types | {'default': 'application/octet-stream'}                               |
| Algorithm     | aes                                                                   |
| Bit length    | 256                                                                   |
| Secret type   | symmetric                                                             |
| Mode          | None                                                                  |
| Expiration    | None                                                                  |
+---------------+-----------------------------------------------------------------------+

C:\Users\Administrator>
```

Section 2: Applied Learning

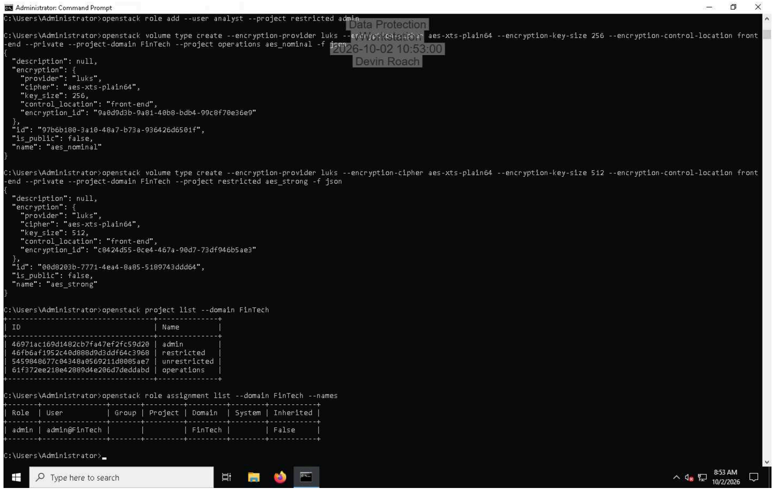
Part 1: Create Scopes for Sensitive Data

10. **Make a screen capture** showing the **project list for the FinTech domain**.



```
| id                  | 2fd9ee9b0ae44216b11bc125b6000303 |
| name                | analyst                          |
| description         | None                             |
| password_expires_at | None                             |
| options             | {}                               |
+---------------------+----------------------------------+

C:\Users\Administrator>openstack role add --user analyst --project restricted admin

C:\Users\Administrator>openstack volume type create --encryption-provider luks --encryption-cipher aes-xts-plain64 --encryption-key-size 256 --encryption-control-location front
-end --private --project-domain FinTech --project operations aes_nominal -f json
{
  "description": null,
  "encryption": {
    "provider": "luks",
    "cipher": "aes-xts-plain64",
    "key_size": 256,
    "control_location": "front-end",
    "encryption_id": "9a0d9d3b-9a81-40b8-bdb4-99c8f70e36e9"
  },
  "id": "97b6b180-3a10-48a7-b73a-936426d6501f",
  "is_public": false,
  "name": "aes_nominal"
}

C:\Users\Administrator>openstack volume type create --encryption-provider luks --encryption-cipher aes-xts-plain64 --encryption-key-size 512 --encryption-control-location front
-end --private --project-domain FinTech --project restricted aes_strong -f json
{
  "description": null,
  "encryption": {
    "provider": "luks",
    "cipher": "aes-xts-plain64",
    "key_size": 512,
    "control_location": "front-end",
    "encryption_id": "c8424d55-0ce4-467a-90d7-73df946b5ae3"
  },
  "id": "00d8203b-7771-4ea4-8a85-5189743ddd64",
  "is_public": false,
  "name": "aes_strong"
}

C:\Users\Administrator>openstack project list --domain FinTech
+----------------------------------+--------------+
| ID                               | Name         |
+----------------------------------+--------------+
| 46971ac169d1482cb7fa47ef2fc59d20 | admin        |
| 46fb6af1952c40d888d9d3ddf64c3968 | restricted   |
| 5459848677c04348a0569211d8085ae7 | unrestricted |
| 61f372ee218e42889d4e206d7deddabd | operations   |
+----------------------------------+--------------+

C:\Users\Administrator>
```

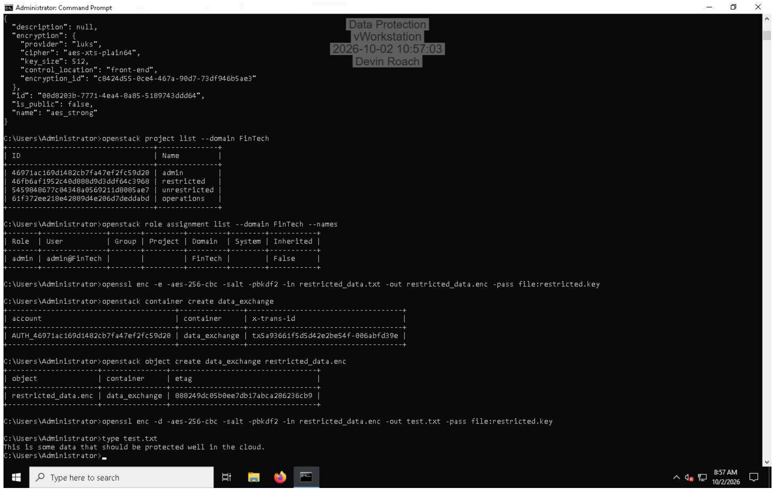
12. **Make a screen capture** showing the **role assignments for the FinTech domain**.



```
C:\Users\Administrator>openstack role add --user analyst --project restricted admin

C:\Users\Administrator>openstack volume type create --encryption-provider luks --encryption-cipher aes-xts-plain64 --encryption-key-size 256 --encryption-control-location front
-end --private --project-domain FinTech --project operations aes_nominal -f json
{
  "description": null,
  "encryption": {
    "provider": "luks",
    "cipher": "aes-xts-plain64",
    "key_size": 256,
    "control_location": "front-end",
    "encryption_id": "9a0d9d3b-9a81-40b8-bdb4-99c8f70e36e9"
  },
  "id": "97b6b180-3a10-48a7-b73a-936426d6501f",
  "is_public": false,
  "name": "aes_nominal"
}

C:\Users\Administrator>openstack volume type create --encryption-provider luks --encryption-cipher aes-xts-plain64 --encryption-key-size 512 --encryption-control-location front
-end --private --project-domain FinTech --project restricted aes_strong -f json
{
  "description": null,
  "encryption": {
    "provider": "luks",
    "cipher": "aes-xts-plain64",
    "key_size": 512,
    "control_location": "front-end",
    "encryption_id": "c8424d55-0ce4-467a-90d7-73df946b5ae3"
  },
  "id": "00d8203b-7771-4ea4-8a85-5189743ddd64",
  "is_public": false,
  "name": "aes_strong"
}

C:\Users\Administrator>openstack project list --domain FinTech
+----------------------------------+--------------+
| ID                               | Name         |
+----------------------------------+--------------+
| 46971ac169d1482cb7fa47ef2fc59d20 | admin        |
| 46fb6af1952c40d888d9d3ddf64c3968 | restricted   |
| 5459848677c04348a0569211d8085ae7 | unrestricted |
| 61f372ee218e42889d4e206d7deddabd | operations   |
+----------------------------------+--------------+

C:\Users\Administrator>openstack role assignment list --domain FinTech --names
+-------+---------------+-------+---------+---------+--------+-----------+
| Role  | User          | Group | Project | Domain  | System | Inherited |
+-------+---------------+-------+---------+---------+--------+-----------+
| admin | admin@FinTech |       |         | FinTech |        | False     |
+-------+---------------+-------+---------+---------+--------+-----------+

C:\Users\Administrator>
```

Part 2: Use Client-Side Encryption

6. **Make a screen capture** showing the **command to see the contents and the result**.



```
{
  "description": null,
  "encryption": {
    "provider": "luks",
    "cipher": "aes-xts-plain64",
    "key_size": 512,
    "control_location": "front-end",
    "encryption_id": "c8424d55-0ce4-467a-90d7-73df946b5ae3"
  },
  "id": "00d8203b-7771-4ea4-8a85-5189743ddd64",
  "is_public": false,
  "name": "aes_strong"
}

C:\Users\Administrator>openstack project list --domain FinTech
+----------------------------------+--------------+
| ID                               | Name         |
+----------------------------------+--------------+
| 46971ac169d1482cb7fa47ef2fc59d20 | admin        |
| 46fb6af1952c40d888d9d3ddf64c3968 | restricted   |
| 5459848677c04348a0569211d8085ae7 | unrestricted |
| 61f372ee218e42889d4e206d7deddabd | operations   |
+----------------------------------+--------------+

C:\Users\Administrator>openstack role assignment list --domain FinTech --names
+-------+---------------+-------+---------+---------+--------+-----------+
| Role  | User          | Group | Project | Domain  | System | Inherited |
+-------+---------------+-------+---------+---------+--------+-----------+
| admin | admin@FinTech |       |         | FinTech |        | False     |
+-------+---------------+-------+---------+---------+--------+-----------+

C:\Users\Administrator>openssl enc -e -aes-256-cbc -salt -pbkdf2 -in restricted_data.txt -out restricted_data.enc -pass file:restricted.key

C:\Users\Administrator>openstack container create data_exchange
+---------------------------------------+---------------+------------------------------------+
| account                               | container     | x-trans-id                         |
+---------------------------------------+---------------+------------------------------------+
| AUTH_46971ac169d1482cb7fa47ef2fc59d20 | data_exchange | tx5a93661f5d5d42e2be54f-006abfd39e |
+---------------------------------------+---------------+------------------------------------+

C:\Users\Administrator>openstack object create data_exchange restricted_data.enc
+---------------------+---------------+----------------------------------+
| object              | container     | etag                             |
+---------------------+---------------+----------------------------------+
| restricted_data.enc | data_exchange | 888249dc05b0ee7db17abca286236cb9 |
+---------------------+---------------+----------------------------------+

C:\Users\Administrator>openssl enc -d -aes-256-cbc -salt -pbkdf2 -in restricted_data.enc -out test.txt -pass file:restricted.key

C:\Users\Administrator>type test.txt
This is some data that should be protected well in the cloud.
C:\Users\Administrator>_
```

Section 3: Challenge and Analysis

Part 1: Investigate Volume Encryption Options

1. **Research** the available encryption options and **make a recommendation** for the mode to use for your most sensitive data.

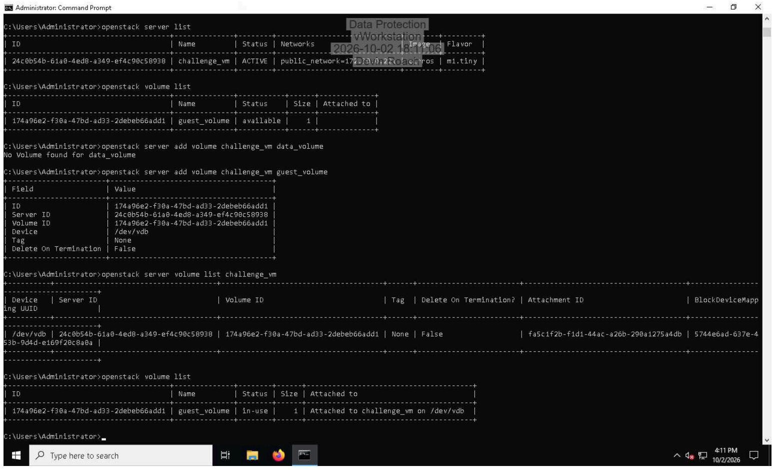
AES-XTS with the plain64 initialization vector mode (aes-xts-plain64) is what I would recommend for the most sensitive data. OpenStack recommends the use of aes-xts-plain64 as the cipher for encrypted volumes, and the other option is aes-cbc-essiv. AES-XTS is a standard for storage device protection and is designed for storage encryption. Similarly, the LUKS/cryptsetup documentation lists aes-xts-plain64 as the default for LUKS and says that, in applicable scenarios, XTS can offer better security than CBC-ESSIV. If the private cloud handles sensitive information, I would opt for aes-xts-plain64 + LUKS. OpenStack documentation shows this configuration using a 256-bit encryption key and front-end encryption.

Part 2: Attach a Volume to a Virtual Machine Instance

1. **Document** the **command to attach a volume**.

openstack server start challenge_vm
openstack server list
openstack volume list
openstack server add volume challenge_vm guest_volume
openstack server volume list challenge_vm
openstack volume list

2. **Make a screen capture** showing a **running VM with a volume attached**.



```
C:\Users\Administrator>openstack server list
+--------------------------------------+--------------+--------+------------------------------+--------+----------+
| ID                                   | Name         | Status | Networks                     | Image  | Flavor   |
+--------------------------------------+--------------+--------+------------------------------+--------+----------+
| 24c0b54b-61a0-4ed8-a349-ef4c90c58938 | challenge_vm | ACTIVE | public_network=172.[illegible] | cirros | m1.tiny |
+--------------------------------------+--------------+--------+------------------------------+--------+----------+

C:\Users\Administrator>openstack volume list
+--------------------------------------+--------------+-----------+------+-------------+
| ID                                   | Name         | Status    | Size | Attached to |
+--------------------------------------+--------------+-----------+------+-------------+
| 174a96e2-f30a-47bd-ad33-2debeb66add1 | guest_volume | available |    1 |             |
+--------------------------------------+--------------+-----------+------+-------------+

C:\Users\Administrator>openstack server add volume challenge_vm data_volume
No Volume found for data_volume

C:\Users\Administrator>openstack server add volume challenge_vm guest_volume
+-----------------------+--------------------------------------+
| Field                 | Value                                |
+-----------------------+--------------------------------------+
| ID                    | 174a96e2-f30a-47bd-ad33-2debeb66add1 |
| Server ID             | 24c0b54b-61a0-4ed8-a349-ef4c90c58938 |
| Volume ID             | 174a96e2-f30a-47bd-ad33-2debeb66add1 |
| Device                | /dev/vdb                             |
| Tag                   | None                                 |
| Delete On Termination | False                                |
+-----------------------+--------------------------------------+

C:\Users\Administrator>openstack server volume list challenge_vm
+----------+--------------------------------------+--------------------------------------+------+-------------------------+--------------------------------------+-----------------
---------------------+
| Device   | Server ID                            | Volume ID                            | Tag  | Delete On Termination? | Attachment ID                        | BlockDeviceMapp
ing UUID             |
+----------+--------------------------------------+--------------------------------------+------+-------------------------+--------------------------------------+-----------------
---------------------+
| /dev/vdb | 24c0b54b-61a0-4ed8-a349-ef4c90c58938 | 174a96e2-f30a-47bd-ad33-2debeb66add1 | None | False                   | fa5c1f2b-f1d1-44ac-a26b-290a1275a4db | 5744e6ad-637e-4
53b-9d4d-e169f20c8a0a |
+----------+--------------------------------------+--------------------------------------+------+-------------------------+--------------------------------------+-----------------
---------------------+

C:\Users\Administrator>openstack volume list
+--------------------------------------+--------------+--------+------+----------------------------------------+
| ID                                   | Name         | Status | Size | Attached to                            |
+--------------------------------------+--------------+--------+------+----------------------------------------+
| 174a96e2-f30a-47bd-ad33-2debeb66add1 | guest_volume | in-use |    1 | Attached to challenge_vm on /dev/vdb  |
+--------------------------------------+--------------+--------+------+----------------------------------------+

C:\Users\Administrator>_
```

Data Protection
vWorkstation
2026-10-02 18:11:06
Devin.Roach